

ACCTG 211H — Mechanics Drill Set (Answer Key)

Week 8 — Long-Term Assets, Depreciation, and Impairment

Drill Set

Required: 1, 2, 5, 6, 7, 8, 9. Due Sunday March 7, 2027, 11:59 p.m. Eastern, after both class meetings. Graded for completion: attempt the required items yourself, check the key, and correct your work. No generative AI. Optional items: 3–4; optional items do not count toward completion or required assessment.

1. (*Capitalized cost.*) A firm buys a machine: invoice price \$50,000, freight-in \$2,000, installation \$3,000, and prepaid insurance for the coming year \$1,500. Compute the cost to be capitalized.
2. (*Straight-line depreciation.*) The machine costs \$55,000, has a \$5,000 salvage value, and a five-year useful life. Compute annual depreciation and the book value at the end of year 2.
3. **Optional.** (*Double-declining-balance.*) For the same machine (cost \$55,000, salvage \$5,000, five-year life), compute depreciation expense for years 1 and 2 under the double-declining-balance method.
4. **Optional.** (*Units-of-production.*) A machine costs \$80,000, has an \$8,000 salvage value, and is expected to produce 180,000 units over its life. It produces 30,000 units in year 1. Compute year-1 depreciation.
5. (*Partial-year depreciation.*) Equipment costing \$24,000 (no salvage, four-year life, straight-line) is purchased and ready for use on October 1. Compute depreciation for the first partial calendar year.
6. (*Disposal.*) Equipment with a cost of \$40,000 and accumulated depreciation of \$30,000 is sold for \$7,000. Compute the gain or loss and record the journal entry.
7. (*Impairment.*) Treat a held-and-used long-lived asset as the entire relevant asset group for this simplified U.S. GAAP case. First compare carrying amount with undiscounted cash flows; if not recoverable, write down to fair value. It has a book value of \$120,000, estimated undiscounted future cash flows of \$110,000, and a fair value of \$95,000. Is the asset impaired, and if so, by how much?
8. (*Goodwill.*) In a simplified acquisition of an entire business (no prior ownership or noncontrolling interest), an acquirer pays \$500,000 for a target whose identifiable net assets have a fair value of \$430,000. Compute goodwill.
9. (*Intangible amortization.*) A patent is acquired for \$60,000 and has a remaining useful life of ten years, no residual value, and straight-line amortization. Compute annual amortization and the carrying value at the end of year 3.

Self-Check Answers

1. **Capitalized cost.** $= 50,000 + 2,000 + 3,000 = \mathbf{\$55,000}$. The prepaid insurance (\$1,500) is a separate asset initially and becomes expense as coverage is consumed; it is not equipment cost.
2. **Straight-line.** Annual depreciation $= (55,000 - 5,000) \div 5 = \$10,000$. Book value end of year 2 $= 55,000 - 2 \times 10,000 = \mathbf{\$35,000}$.
3. **Optional — Double-declining-balance.** Rate $= 2 \div 5 = 40\%$. Year 1 $= 55,000 \times 40\% = \mathbf{\$22,000}$. Year 2 $= (55,000 - 22,000) \times 40\% = 33,000 \times 40\% = \mathbf{\$13,200}$. (Do not depreciate below the salvage value.)
4. **Optional — Units-of-production.** Rate $= (80,000 - 8,000) \div 180,000 = \0.40 per unit. Year 1 $= 30,000 \times 0.40 = \mathbf{\$12,000}$.
5. **Partial year.** Full-year depreciation $= 24,000 \div 4 = \$6,000$. October 1 to December 31 = 3 months: $6,000 \times 3/12 = \mathbf{\$1,500}$.
6. **Disposal.** Book value $= 40,000 - 30,000 = \$10,000$. Proceeds $\$7,000 < \$10,000$, so **loss of \$3,000**. Entry: Dr Cash \$7,000; Dr Accumulated Depreciation \$30,000; Dr Loss on Disposal \$3,000; Cr Equipment \$40,000.
7. **Impairment.** Recoverability test: undiscounted cash flows $\$110,000 < \text{book value } \$120,000$, so the asset **is impaired**. Impairment loss $= \text{book value} - \text{fair value} = 120,000 - 95,000 = \mathbf{\$25,000}$.
8. **Goodwill.** $= 500,000 - 430,000 = \mathbf{\$70,000}$.
9. **Patent amortization.** Annual $= 60,000 \div 10 = \$6,000$. Carrying value end of year 3 $= 60,000 - 3 \times 6,000 = \mathbf{\$42,000}$.